

Do our scripted (MP) results follow Sokolić et al., Dinh et al., and Rahaman et al.?

A pre-specified test suite over three papers, with verdicts and the cells that failed

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Scope. “Scripted data” means the MP collector only (SE(3) min-jerk plan, feedforward/feedback tracking, reactive contact handoff); the pointing-label dataset is retired. Six trained arms: {L2, HT, HG, MIP} at 200 demos and {L2, MIP} at 2000 demos, all 300k steps, one seed each. Success rate (SR) is the twofactor harness, AS= 8, 60 seeds. All Jacobians below are of the *deployed* policy map (observation window $\rightarrow$ executed 8-step action chunk, through the full sampler) unless stated otherwise.

The two theories. (A) Sokolić, Giryes, Sapiro, Rodrigues, *Robust Large Margin Deep Neural Networks* (arXiv:1605.08254): the generalization error is controlled by the *spectral norm of the network Jacobian in the neighbourhood of the training samples* (divided by the classification margin), with no dependence on width or depth; they also propose a Jacobian-norm regularizer. (B) Dinh, Pascanu, Bengio, Bengio, *Sharp Minima Can Generalize For Deep Nets* (arXiv:1703.04933): for rectifier networks, parameter-space flatness/sharpness measures can be changed arbitrarily by function-preserving reparameterizations, so they cannot by themselves explain generalization; any meaningful measure must be invariant.

1 Theory A: the Jacobian spectral norm

A1 — cross-arm test

Two independent measurements were made. Both use the deployed map, but the first drew states and gaps from separate probes; the second (`probe_theory_suite`) measures $\|J\|_2$, the train error and the held-out error in *one* pass with a single state protocol. We report the second as primary.

arm	$\ J\ _2$	N_{win}	$\ J\ _2/\sqrt{N}$	gap	held-out err	SR
MIP-2k	3.86	74111	0.0142	0.00428	0.00464	100
L2-2k	4.24	74111	0.0156	0.00433	0.00546	94
HT-200	2.18	7431	0.0253	0.01169	0.01179	75
MIP-200	2.38	7431	0.0276	0.01198	0.01211	94
L2-200	2.99	7431	0.0347	0.01141	0.01192	63
HG-200	3.08	7431	0.0357	0.01084	0.01107	87

$r(\|J\|_2/\sqrt{N}, \text{gap}) = +0.857$ and $r(\|J\|_2/\sqrt{N}, \text{held-out error}) = +0.869$ — but the correlation is carried *entirely by the data-scale contrast*: the two 2k arms have small gaps and small $\|J\|_2/\sqrt{N}$, the four 200-demo arms have large ones. **Within the 200-demo group the relation vanishes**: sorted by $\|J\|_2$ the gaps read $2.18 \rightarrow 0.0117$, $2.38 \rightarrow 0.0120$, $2.99 \rightarrow 0.0114$, $3.08 \rightarrow 0.0108$ — i.e. flat to slightly *decreasing*. The apparent cross-arm success of Theory A is a two-group effect that $\sqrt{N}$ already accounts for. **refuted** as an explanation of the objective axis; **partial** as a description of the data axis.

Two recorded corrections. (i) A first pass using Frobenius norms at rollout states gave $r = 0.966$ and a 6/6 monotone ordering; it does not survive the single-protocol measurement. (ii) In that first pass MIP-200’s gap (0.0073) was below L2-200’s (0.0095); under the controlled protocol the ordering *reverses* (0.0120 vs 0.0114). The between-arm gap differences at 200 demos are within the state-sampling noise, so no claim should rest on them.

A2 — within-arm trajectory (no cross-arm confound)

Tracking both quantities across snapshots of the *same* arm removes every cross-arm confound. The spectral norm falls by a factor of 2.7–4.7 over training in every arm; the gap does not follow it.

snapshot	20k	60k	120k	180k	300k
L2-200 $\ J\ _2$	7.93	5.15	3.14	2.94	2.99
L2-200 gap	0.0103	0.0127	0.0090	0.0101	0.0114
MIP-200 $\ J\ _2$	7.38	2.51	2.40	2.37	2.38
MIP-200 gap	0.0089	0.0092	0.0136	0.0125	0.0120
L2-2k $\ J\ _2$	13.76	13.05	5.40	4.61	4.24
L2-2k gap	0.0027	0.0038	0.0034	0.0035	0.0043

In L2-200 the norm falls $2.7\times$ while the gap rises slightly; in MIP-200 it falls $3.1\times$ while the gap rises 35%; in L2-2k it falls $3.2\times$ while the gap rises 60%. **refuted** — within a training run the Jacobian spectral norm and the generalization gap move in opposite directions.

A3 — their own regularizer, correctly dosed

Sokolić et al. propose penalising the Jacobian norm. Our first attempt ($\lambda = 10^{-3}$) destroyed the fit (SR 2/60, on-support error +35%) and is a void cell. Two correctly-dosed arms ($\lambda = 10^{-5}, 10^{-4}$) are training; the prediction is a *lower* $\|J\|_2$ and a *smaller* gap, with the fit intact.

A4 — width independence

The bound is claimed independent of width. A $2\times$ -width L2 arm is training; the prediction is that its gap follows its measured $\|J\|_2$, not its parameter count.

2 Theory B: reparameterization invariance

The encoder is an MLP with LeakyReLU, which is positively homogeneous, so scaling the units of a hidden layer by α and dividing the next layer’s incoming weights by α leaves the function *exactly* unchanged. Applied to L2-200 (measured function change: mean 3.5×10^{-5} , on an action scale of 0.62 — float error):

statistic	before	after	verdict
encoder weight stable rank	[18.8, 21.0, 21.1]	[4.7, 7.9, 21.1]	gauge artifact
embedding covariance PR	3.246	3.247	invariant
encoder-Jacobian PR	4.609	4.609	invariant
deployed-map PR	1.9460	1.9461	invariant
deployed $\ J\ _2$	2.9809	2.9809	invariant

Consequence for our own evidence. We had cited the encoder weight stable rank as data-free evidence that the L2/MIP split “lives in the weights”. That claim is **refuted** and is withdrawn: the same function admits stable ranks of 18.8 and 4.7.

What survives, and why. The embedding PR survives this symmetry because the conditioning path applies a pointwise GELU *directly* to the embedding (`cond_encoder = Sequential(GELU, Linear)`), which pins the embedding’s coordinate system — it is not a free gauge in this architecture. It is, however, only weakly protected: rescaling the embedding’s output units and compensating in the (GELU-preceded) consumers moved `emb_PR` from 3.25 to 4.84 (+49%) while changing the function by $\sim 1\%$. We therefore treat `emb_PR` as admissible but flagged, and treat the deployed-map quantities — which are function-space objects and invariant by construction — as the primary evidence.

B2 — ϵ -sharpness

Dinh’s own quantity $S(\theta) = \max_{\|\delta\| \leq \epsilon \|\theta\|} (L(\theta + \delta) - L(\theta)) / (1 + L(\theta))$, $\epsilon = 5 \times 10^{-4}$, 16 random directions, measured before and after the *exact* function-preserving rescaling:

arm	$S(\theta)$	after rescaling	ratio
L2-200	1.93×10^{-3}	5.19×10^{-4}	$0.27\times$
L2-2k	3.95×10^{-3}	1.35×10^{-3}	$0.34\times$
HT-200	1.50×10^{-2}	2.55×10^{-2}	$1.70\times$
HG-200	1.19×10^{-1}	3.26×10^{-2}	$0.27\times$
MIP-200	6.25×10^{-2}	8.09×10^{-2}	$1.29\times$
MIP-2k	6.95×10^{-2}	0	$0\times$

Sharpness moves by factors of 0.27 to 1.7 (and to exactly zero for MIP-2k) under a transformation that leaves every prediction unchanged. **supported** — Dinh’s construction applies to our networks directly.

Two further observations. First, the deployed $\|J\|_2$ and deployed PR are invariant to *machine precision* under both the rescaling and an exact hidden-unit permutation (e.g. L2-200: $2.9903 \rightarrow 2.9903 \rightarrow 2.9903$), confirming the function-space/parameter-space split. Second, and directly in the spirit of the paper’s title: MIP-200 sits in a minimum $32\times$ *sharper* than L2-200’s (6.3×10^{-2} vs 1.9×10^{-3}) while generalizing at least as well and succeeding 50% more often. Sharp minima generalize here too. (Caveat: the loss functions differ between arms, so cross-arm sharpness comparisons are suggestive, not rigorous; the within-arm reparameterization results are the rigorous part.)

3 Theory C: spectral bias (Rahaman et al., arXiv:1806.08734)

Claim. ReLU networks learn low frequencies first and express high frequencies only with finely tuned parameters, so high-frequency content is acquired late and is fragile.

Test. A demonstration trajectory is a dense 1-D path through observation space with ground-truth actions at every point, so the frequency content of a policy’s output along that path can be compared directly with the target’s own spectrum. Over 12 trajectories per arm we report the captured power ratio (pred/GT) in three bands ($f < 0.05$, $0.05 \leq f < 0.2$, $f \geq 0.2$ cycles/step), the error power by band, and the high-band power lost under a small parameter perturbation. The target itself is strongly low-frequency: its high/low power ratio is 0.0102.

arm	captured high-band power ratio			err. frac.	frag.
	20k	60k	300k	high @300k	high
L2-200	0.901	0.913	0.996	0.587	+0.001
MIP-200	0.978	0.993	0.988	0.634	−0.004
HT-200	0.919	0.958	0.988	0.624	−0.009
HG-200	0.752	0.891	0.984	0.728	+0.001
L2-2k	0.723	0.860	0.993	0.383	+0.001
MIP-2k	1.012	0.964	1.000	0.352	+0.000

Learning order: supported. High-band capture rises monotonically with training in four of six arms (HG $0.75 \rightarrow 0.98$, L2-2k $0.72 \rightarrow 0.99$), i.e. low frequencies are fitted first — exactly the predicted order. The residual error is high-frequency dominated in every arm (35–73% of error power above $f = 0.2$), again as predicted.

A new objective difference. MIP acquires the high-frequency content first: at 20k steps it is already at 0.978 (200 demos) and 1.012 (2k) while HG is at 0.752 and L2-2k at 0.723. This lines up with two independent measurements: MIP rebuilds representation rank from ~ 1000 steps (§B), and MIP’s success rate peaks at 20–60k. Fast acquisition of fine structure is a genuine, measurable property of the objective.

Endpoint deficiency: refuted. By 300k *every* arm captures 98–100% of the target’s high-band power. The SR spread of 63–100 therefore cannot be attributed to missing high-frequency content along the demonstration manifold.

Fragility: not observed. A relative parameter perturbation of 2×10^{-3} changes high-band power by less than 1% in every arm ($|\text{frag}| \leq 0.009$), so the “finely tuned parameters” claim does not manifest at this scale in our networks.

4 What the theories do and do not explain

Explained. Theory C’s learning-order claim is confirmed, and it yields a new and useful fact: MIP reaches the target’s high-frequency content several times faster than the other objectives, which coincides with its early SR peak. Theory B is confirmed outright and gives us a measurement filter we now apply: *function-space quantities only*. It also retires two things we had been quoting — encoder weight stable rank (a gauge artifact) and any prospective “MIP finds flatter minima” story (MIP’s minima are $32\times$ sharper). Theory A survives only as a statement about the *data* axis: $\|J\|_2/\sqrt{N}$ separates the 2k arms from the 200-demo arms in the right direction.

Not explained. (i) The objective axis: at matched data the four objectives span 63–94 SR with $\|J\|_2$ spanning only 2.18–3.08 in the *wrong* order (HT lowest norm, second-worst SR; HG highest norm, third-best SR) and with held-out errors that agree to within sampling noise (0.0111–0.0121). Neither theory addresses why these four functions, which are equally accurate off their training sets, behave so differently in closed loop. (ii) The *formation* asymmetry: from an identical initialization (emb_PR 4.24, encoder-Jacobian PR 35.7), every objective collapses to emb_PR ≈ 2.6 –3.1 within 25–100 steps, after which HG/HT rebuild from ~ 400 steps, MIP from ~ 1000 , and L2 barely rebuilds at all ($2.65 \rightarrow 3.69$ by 20k versus HG’s $2.61 \rightarrow 6.54$). Neither theory addresses which objectives rebuild.

Interventions attempted (honest accounting). Five arms designed to move a candidate property have been damaged by dose or by train/test mismatch and are void cells, not evidence: obs-noise ($\sigma = 0.05$, SR 12), jacreg ($\lambda = 10^{-3}$, SR 2), condreg v1 ($\lambda = 1$, SR 0), featdrop $p = 0.05$ (SR 17) and $p = 0.15$ (SR 2). The featdrop pair is instructive nonetheless: it *did* raise emb_PR ($3.29 \rightarrow 4.23$) and it *did* shrink the gap ($0.0095 \rightarrow 0.0063$), while SR collapsed — because the level of the held-out error rose ($0.0100 \rightarrow 0.0136$). SR follows the error level, not the gap. Correctly-dosed replacements (condreg v2 at 3×10^{-3} and 3×10^{-4} , jacreg at 10^{-5} and 10^{-4}) are training.

5 The experiments run to try to establish the account

Every measurement below was undertaken specifically to give the three theories the best chance of explaining MIP/flow’s advantage.

E1 — Sokolić where the margin is tight (insertion phase)

The bound is $\text{margin}/\|J\|_2$; our margin is smallest at insertion, so that is where the theory should bite. A first pass ($n = 40$ states) looked decisive — MIP’s insertion-phase gain appeared $3\times$ below L2’s at both data scales (5.6 vs 16.2; 10.3 vs 35.6). **It does not replicate at $n = 120$:** the means are outlier-dominated and unstable (MIP-200 now reads 56.3, *above* L2-200’s 26.3). Robust statistics give

arm	gain median	gain p90	SR
MIP-200	2.40	18.6	94
HT-200	2.71	54.4	75
HG-200	3.11	67.4	87
L2-200	3.11	49.4	63
L2-2k	4.10	74.2	94
MIP-2k	4.47	23.7	100

$r = +0.18$ (median) and -0.02 (p90) against SR. The one real signal is that *both MIP arms have far fewer high-gain insertion states* (p90 18.6/23.7 versus 49–74 for every other arm) — an objective signature, but one that does not order the remaining four arms. **refuted** as stated; **partial** as a MIP-vs-rest contrast.

E2 — Rahaman off the manifold (pre-registered falsifier)

On-manifold every arm is spectrally complete, so the pre-registered test was whether fine structure survives *off* it. Along captured off-support rollout states ($d > 10$, 600 states) versus on-support ones:

arm	high-band frac. on	off	output power off/on	SR
MIP-200	0.607	0.559	0.29	94
MIP-2k	0.610	0.597	0.30	100
L2-2k	0.610	0.605	0.49	94
HT-200	0.606	0.586	0.57	75
HG-200	0.608	0.582	0.71	87
L2-200	0.606	0.590	1.03	63

The spectral *shape* is preserved by every arm off-manifold (0.56–0.60 high-band fraction, essentially unchanged). What differs is the *total output power*: the MIP arms attenuate to 29–30% of their on-manifold energy, L2-2k to 49%, and L2-200 not at all (103%). This orders SR 5/6 with $r = -0.844$ — the best of any quantity in this report — but it is a statement about response *magnitude* off the data, i.e. a re-measurement of the off-support behaviour we already had ($|a|$ far-field 0.40 for MIP vs 0.79 for L2), not a spectral-bias effect.

E3 — causal tests of the in-support account

The incoherent (within-chunk) component of the held-out error orders SR 6/6 — the only on-manifold quantity that separates HT from MIP. Two manipulations tested it causally. Injecting incoherent error into MIP-200 at doses spanning the entire between-arm range (0.05–0.40mm, both i.i.d. and the faithful state-correlated zero-mean-per-chunk version) left SR unchanged: 52/60 $\rightarrow$ 51, 54, 50, 51, 52, 57. Removing it from L2-200 by within-chunk low-pass destroyed the policy (the filter attenuates the coherent component too) and is void. **refuted** — the correlation does not survive intervention; the PD/impedance layer and arm inertia evidently absorb incoherent action error at these magnitudes.

E4 — ablations

Batch size 256 vs 1024: SR 41/60, 38/60 vs 40/60, 38/60; in-support error identical (1.054 vs 1.046mm). The gradient-noise knob does nothing, weakening every noise-driven account. Coverage: 200 $\rightarrow$ 38/60, 800 $\rightarrow$ 55/60, 2000 $\rightarrow$ 56/60, with in-support error falling 1.046 $\rightarrow$ 0.754 $\rightarrow$ 0.449mm — on the *data* axis precision and SR move together, which is what an on-manifold theory predicts and is the one place the theories work.

6 Verdict

theory	verdict	what it does and does not carry
Dinh	supported	Applies to our networks exactly; retires weight stable rank and any flat-minima story (MIP’s minima are $32\times$ sharper); supplies the function-space measurement filter. Methodological, not explanatory.
Sokolić	partial	Describes the <i>data</i> axis (2k vs 200). Fails within a training run (norm falls $2.7\text{--}4.7\times$ while the gap is flat or rises), fails within matched data, and its margin-weighted form fails at adequate n .
Rahaman	partial	Learning order confirmed, and MIP acquires high-frequency content $\sim 3\times$ earlier — a real objective difference. But every arm is spectrally complete by 300k, so it cannot explain the endpoint spread.

The honest conclusion. The three papers are theories of on-manifold supervised learning, and on that ground our six arms differ in the ways the papers describe — but those differences do not survive intervention, while the differences that track SR (off-support response magnitude, $r = -0.84$; excursion escalation, 0.44 vs

0.12; the restoring-versus-escaping annulus field) are all off-manifold quantities that none of the three papers addresses. We therefore cannot claim these papers explain MIP/flow’s benefit. What we can claim is narrower and defensible: *Dinh tells us which of our measurements are admissible; Sokolić accounts for why more data helps; Rahaman accounts for why MIP gets there first; and the residual advantage — the part that separates the objectives at fixed data — is an off-manifold property that remains unexplained by all three.*

7 Pre-registered falsifiers

1. If the within-arm trajectory (A2) shows the gap and $\|J\|_2$ moving in opposite directions over training, Theory A does not describe our training dynamics, only our cross-arm variation.
2. If correctly-dosed jacreg (A3) lowers $\|J\|_2$ with the fit intact and the gap does *not* shrink, Theory A’s mechanism is refuted constructively in our setting.
3. If the $2\times$ -width arm (A4) has the same $\|J\|_2$ and a different gap, the width-independence claim fails here.
4. If ϵ -sharpness is unchanged by the exact rescaling (B2), our architecture escapes Dinh’s construction and sharpness would become admissible again.
5. If a frequency-resolved measurement *off* the demonstration manifold (the regime where the arms actually differ) shows the spectral deficiency that the on-manifold measurement does not, Theory C would be reinstated as an explanation of the objective axis rather than only of the learning order.